Master of Data Science & Artificial Intelligence

CS5998 | Capstone Project

Milestone 3: Final Report

Data-Driven Power System Upgrade Planning for Cellular Network Sites Using Traffic & Hardware Configuration

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Milestone 3 – Final Report

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1. Introduction

1.1. Background

The rapid evolution of mobile communication technologies has significantly increased the energy demand and complexity of modern telecommunication networks. Operators continuously upgrade Radio Access Network (RAN) infrastructure by deploying LTE and 5G equipment such as RRUs, AAUs, and baseband units to improve coverage, capacity, and Quality of Service (QoS). These upgrades substantially increase power consumption compared to traditional 2G and 3G systems.

Modern telecom sites typically support multiple technologies simultaneously, making site-level power demand highly dynamic and difficult to estimate using conventional engineering calculations. Site power systems, including rectifiers, battery banks, generators, and cooling systems, may become overloaded if upgrades are introduced without proper power planning, potentially causing service outages and infrastructure failures.

Traditionally, operators rely on vendor specifications and rule-of-thumb engineering methods to estimate power requirements. However, these approaches often fail to capture the relationship between traffic behavior, hardware utilization, and actual site power consumption, especially in large multi-technology networks.

Meanwhile, telecom operators maintain large operational datasets containing traffic statistics, hardware inventory information, and historical power measurements. These datasets enable the development of data-driven power prediction models capable of improving infrastructure planning accuracy.

This capstone project focuses on predicting telecom site power consumption using operational data obtained from Cable & Wireless Seychelles (C&WS). The project compares traditional engineering-rule methods with multiple machine learning approaches to identify the most accurate and operationally suitable framework for power upgrade planning. Although the original scope included broader analytics capabilities, the final implementation concentrated primarily on accurate site-level power prediction.

1.2. Motivation

Accurate prediction of telecommunication site power consumption is critical for mobile network operators, especially during LTE and 5G infrastructure upgrades. Incorrect estimation of electrical demand can create serious operational and financial risks.

Under-provisioned power systems may lead to rectifier overload, battery degradation, and complete service outages affecting multiple services and shared infrastructure. Such failures can result in customer dissatisfaction, SLA penalties, emergency maintenance, and revenue loss. The challenge becomes more severe for remote or non-motorable sites where restoration activities are costly and time-consuming.

Conversely, over-provisioning increases unnecessary CAPEX through oversized rectifiers, batteries, and power infrastructure. In large-scale upgrade programs, even small estimation inaccuracies may lead to significant financial inefficiencies.

Despite these risks, many operators still rely on static engineering calculations and vendor specifications for power planning. While useful for approximate estimation, these methods often fail to capture the dynamic relationship between traffic load, hardware configuration, and actual site power consumption, particularly in modern multi-technology networks.

Electrical power consumption at telecommunication sites is strongly influenced by the traffic generated by individual sectors and cells. This relationship provides a strong foundation for developing a data-driven power system upgrade planning framework for cellular network sites using traffic and hardware configuration data obtained from a leading telecommunication service provider in Seychelles.

1.3. Traffic Distribution of a 5G NR Cell Vs Time

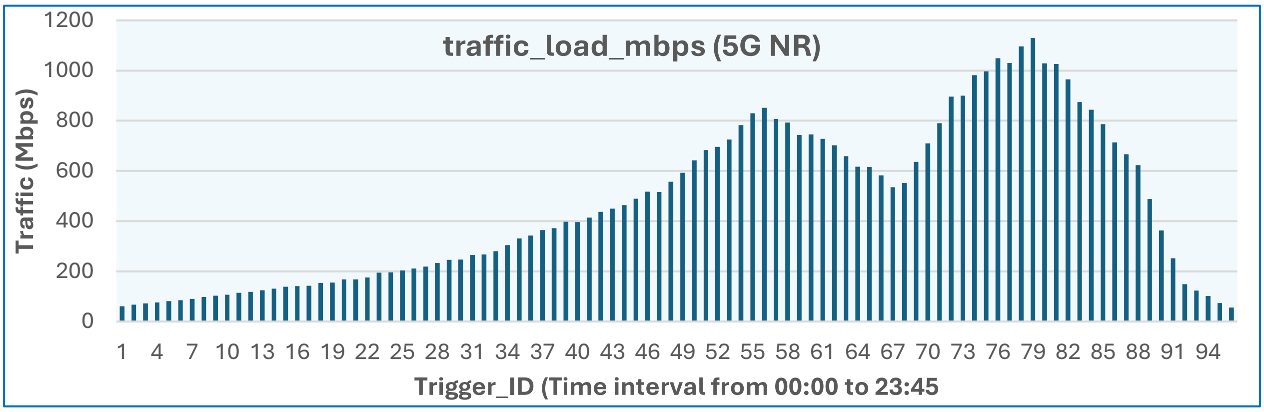

Figure 1-1: 5G Traffic Vs Time

1.4. Traffic Distribution of a 4G LTE Cell Vs Time

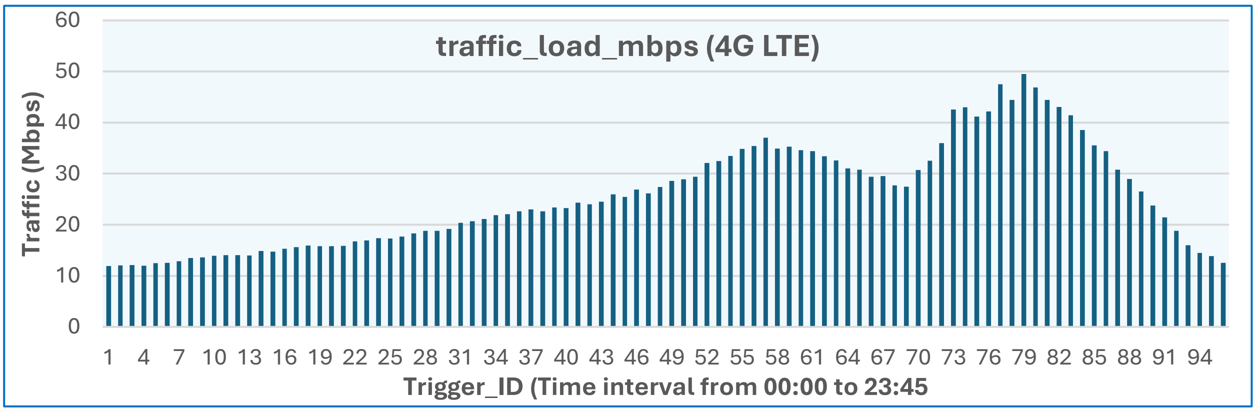

Figure 1-2: 4G Traffic Vs Time

1.5. Average Traffic Profile (NR vs LTE)

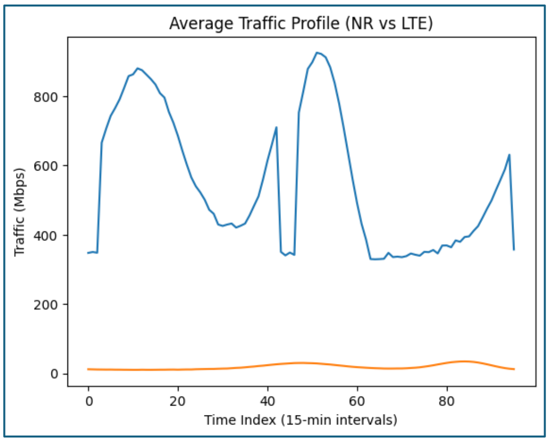

Figure 1-3: Average Traffic Profile (NR vs LTE)

Note:
Maximum traffic level served by 5G is 1,300 Mbps and whereas 4G can serve 80 Mbps maximum.

1.6. Site Traffic Vs Total Power Consumption

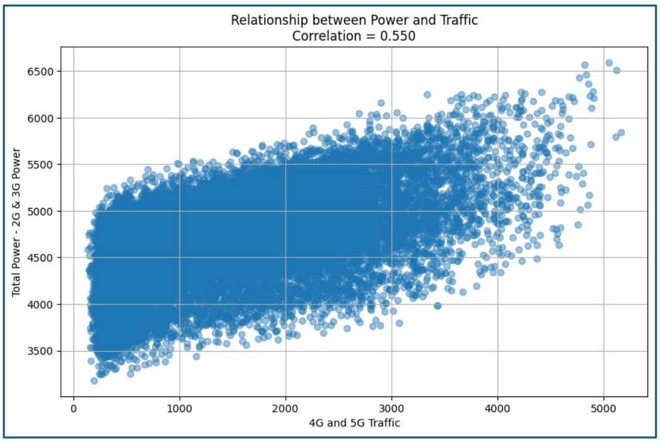

Figure 1-4: NR Traffic vs Power

Note:
Power Consumption caused by 2G and 3G RRUs are removed, Considered the sites with Power greater than 2000 W.

Machine learning techniques provide an opportunity to improve prediction accuracy by learning complex relationships directly from historical operational data. By integrating traffic telemetry, hardware inventory data, and measured site power consumption, ML-based frameworks can generate more adaptive and scalable power prediction models suitable for modern telecommunication environments.

This project was therefore motivated by the need to:

- improve the accuracy of telecommunication site electrical power consumption prediction
- reduce operational and financial risk during infrastructure upgrades
- evaluate the effectiveness of engineering-rule versus data-driven approaches
- develop a practical decision-support framework for telecommunication power planning teams

1.7. Objectives of the Capstone Project

The primary objective of this project is to develop and evaluate a data-driven decision-support framework for predicting power consumption of telecommunication sites undergoing hardware and traffic growth changes.

The specific objectives of the project are as follows:

- To develop multiple machine learning frameworks capable of predicting site-level power consumption using traffic and hardware configuration data.
- Comparing cell-wise and site-wise machine learning methodologies for telecom power prediction.
- To evaluate model performance using quantitative regression metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2).
- To develop a rule-based engineering framework for estimating power consumption of multi-technology telecommunication sites.
- To identify the operational advantages and limitations of engineering-rule and machine learning approaches.
- To propose a scalable framework suitable for supporting telecommunication power systems and upgrade planning activities.

1.8. Scope of the Capstone Project

This project focuses on power prediction for cellular network sites operated by Cable & Wireless Seychelles (C&WS). The analysis primarily utilizes LTE traffic data, 5G NR traffic data, site-level power measurements and site hardware configuration information.

The dataset consists of total power supplied by the AC-DC rectifier system, recorded at 15-minute intervals (default system measurement interval), along with corresponding 4G and 5G traffic levels (Mbps) over a period of seven consecutive days.

The data was collected from all major sites deployed by C&W Seychelles to provide nationwide network coverage.

The project evaluates four different power prediction methodologies:

- Engineering-rule-based prediction
- Fully data-driven cell-wise ML prediction
- Fully data-driven site-wise ML prediction
- Final hybrid ML prediction framework

The project scope is limited to power prediction and comparative analysis. Advanced functionalities such as long-term traffic forecasting, anomaly detection, clustering, and live deployment dashboards were identified during project planning but were not fully implemented within the final project timeline.

1.9. Capstone Project Contributions

This project makes several practical and technical contributions to the field of telecommunication in terms of estimating electrical power consumption of sites planned for RAN and other service upgrades.

1.9.1. Contribution 1 — Fully Data-Driven Cell-Wise ML Framework

A machine learning framework was developed to estimate cell-level power consumption using traffic information before aggregating predictions to site level. This approach enabled evaluation of decomposition-based telecom power prediction.

1.9.2. Contribution 2 — Fully Data-Driven Site-Wise ML Framework

A second machine learning framework was developed using aggregated site-level traffic directly as model input. This approach significantly improved prediction accuracy by reducing aggregation-related error propagation.

1.9.3. Contribution 3 — Engineering-Rule Power Prediction Framework

A telecom-specific engineering-rule framework was developed using technology-specific hardware parameters for 2G, 3G 4G and 5G. The framework estimates total site power consumption using telecom engineering assumptions similar to operational planning practices used in industry.

1.9.4. Contribution 4 — Final Hybrid ML Framework

A final machine learning framework combining traffic information and hardware configuration features was developed using Random Forest regression. This model achieved the best overall performance among all evaluated methods.

1.9.5. Contribution 5 — Comparative Evaluation of Four Prediction Approaches

A detailed comparative analysis was conducted across four different methodologies using MAE, RMSE, MAPE, R^2, scatter plots, site-wise error analysis, feature importance analysis, etc.

The project demonstrates that direct site-level machine learning approaches significantly outperform cell-level decomposition methods for telecommunication power prediction.

2. Problem Statement

2.1. Overview of the Operational Problem

Modern telecommunication networks are experiencing rapid growth in both traffic demand and infrastructure complexity. Mobile network operators continuously expand and modernize their Radio Access Network (RAN) infrastructure to support increasing customer demand for voice, data, video streaming, cloud applications, and Internet of Things (IoT) services. The deployment of Long-Term Evolution (LTE) and Fifth Generation (5G) technologies has significantly increased the processing capability, transmission capacity, and power requirements of cellular network sites.

To improve coverage, throughput, and Quality of Service (QoS), operators regularly introduce hardware upgrades at existing cellular sites. These upgrades may include the installation of additional Radio Remote Units (RRUs), Active Antenna Units (AAUs), baseband processing equipment, microwave transmission systems, and supporting indoor infrastructure.

Although such upgrades improve network performance and customer experience, they simultaneously increase electrical power demand at the site level.

Telecommunication sites are typically powered using integrated DC power systems consisting of rectifiers, battery banks, power distribution units, backup generators, and cooling systems. These systems are designed to support the expected electrical load of all active network equipment under both normal and backup operating conditions.

When new telecommunication hardware is installed without corresponding electrical infrastructure upgrades, the site power system may become overloaded, leading to serious operational risks.

In many practical situations, operators continue to estimate power requirements using static engineering calculations and vendor datasheets. These calculations are often based on fixed assumptions regarding equipment power consumption and average utilization levels. However, modern telecommunication traffic patterns are highly dynamic and vary significantly depending on:

- subscriber density
- time of day
- service usage behavior
- geographical location
- technology generation
- network congestion conditions

As a result, traditional rule-based calculations frequently fail to capture the real operational behavior of modern multi-technology cellular sites. Cable & Wireless Seychelles (C&WS), similar to many global operators, maintains a large operational data lake containing:

- high-resolution traffic telemetry
- site-level power measurements
- hardware inventory records
- configuration data
- historical operational information

This project addresses the challenge of predicting telecommunication site power consumption using both engineering-rule approaches and data-driven machine learning approaches. The project specifically focuses on evaluating how accurately different methodologies can estimate

- site-level power demand using traffic information
- hardware configuration
- technology-specific operational data

The final objective is to support infrastructure planning teams in making more accurate and data-driven decisions regarding power system upgrades.

2.2. Existing industry practices and their limitations

Power planning activities within telecommunication networks are traditionally performed using engineering calculations, vendor specifications, and operational experience. During hardware upgrade projects, planning engineers estimate the expected electrical demand of newly installed equipment based on predefined power ratings provided by equipment manufacturers. For example,

- 2G and 3G RRUs are often assigned fixed nominal power values
- LTE RRUs are estimated using static transmission assumptions
- 5G AAUs are calculated using vendor maximum consumption ratings
- baseband processing units are assigned fixed operational power levels

The estimated power values are then aggregated to determine the total expected power demand of the site. Based on this estimation, engineers decide whether,

- the existing rectifier capacity is sufficient
- battery bank upgrades are required
- additional power modules are needed
- generator systems must be upgraded
- air-conditioning systems need enhancement

Although this engineering-based methodology has been widely used in industry for many years, it suffers from several limitations.

- Traditional engineering calculations assume relatively static equipment behavior. Therefore, actual power consumption may vary significantly over time.
- Engineering-rule methods typically rely on conservative assumptions in order to avoid under-provisioning risks. While this improves safety margins, it may also result in excessive over-provisioning of power infrastructure, increasing capital expenditure unnecessarily.
- Manual engineering calculations have become increasingly difficult to manage in large-scale nationwide modernization programs involving hundreds of sites and multiple technologies.

2.3. Risks of Under-Provisioning

Under-provisioning occurs when the installed electrical infrastructure of a telecommunication site is insufficient to support the actual power demand generated by operational equipment. This represents one of the most critical operational risks in mobile network infrastructure management.

When site-level electrical demand exceeds the capacity of rectifiers or power distribution systems, multiple types of failures may occur simultaneously. Overloaded rectifiers may enter protection mode, reduce output performance, or fail completely. In severe cases, the entire DC power system may shut down in order to protect itself from thermal or electrical damage.

A complete site power shutdown can simultaneously interrupt:

- 2G, 3G, 4G, 5G services
- microwave transmission systems
- monitoring and alarm systems
- other stakeholders' equipment

Such outages directly affect customer experience, network availability, and operator reputation.

Battery systems are also heavily affected by power overload conditions. Rectifiers operating continuously near maximum capacity may fail to properly recharge battery banks. This reduces battery lifetime, accelerates degradation, and compromises backup autonomy during commercial power failures.

Furthermore, delayed power upgrades may postpone the deployment of newly purchased telecommunication equipment. Operators may invest millions of dollars in LTE or 5G hardware but remain unable to activate services until supporting power infrastructure upgrades are completed. This directly affects revenue generation, network expansion timelines, competitive positioning, ROI, etc.

2.4. Risks of Over-Provisioning

Although under-provisioning creates serious operational risks, excessive over-provisioning also introduces significant financial and infrastructure inefficiencies.

In many practical situations, engineers intentionally apply conservative safety margins when estimating future site power demand. While this reduces the probability of overload, it may also lead to the installation of oversized:

- rectifiers
- battery banks
- generators
- cables
- power distribution systems
- air-conditioning units

Such oversized infrastructure significantly increases Capital Expenditure (CAPEX), particularly during nationwide upgrade programs involving hundreds of sites.

Battery systems represent a major cost component of telecommunication power infrastructure. Installing unnecessarily large battery banks increases procurement cost, transportation cost, installation complexity, and maintenance overhead.

Furthermore, modern telecommunication operators face increasing pressure to improve energy efficiency and reduce carbon emissions. Oversized infrastructure conflicts with sustainability objectives by increasing unnecessary energy consumption and resource utilization.

Therefore, power prediction frameworks must balance:

- operational reliability
- infrastructure safety
- financial efficiency
- deployment scalability

2.5. Need for a Data-Driven Decision Support Framework

The increasing complexity of modern telecommunication networks has created a strong need for intelligent and data-driven infrastructure planning systems. Traditional manual calculations are no longer sufficient to accurately model the nonlinear relationship between traffic demand, environmental conditions, technology mix, hardware utilization and actual site power consumption, etc.

At the same time, telecommunication operators already possess large quantities of operational data generated by network management systems, power monitoring systems, and traffic analytics platforms. These datasets provide valuable information regarding real-world network behavior and equipment performance.

Machine learning techniques are capable of identifying hidden relationships within large operational datasets and generating predictive models that continuously adapt to observed data patterns. Unlike traditional engineering calculations, machine learning models can learn nonlinear relationships, adapt to varying traffic conditions, identify hidden operational trends and improve prediction accuracy over time.

Additionally, ML-based systems can process large numbers of sites simultaneously, enabling scalable nationwide analysis that would be extremely difficult using purely manual engineering methods. A data-driven decision support framework can therefore help telecommunication operators:

- improve infrastructure planning accuracy
- reduce outage risk
- optimize capital expenditure
- accelerate hardware deployment
- improve operational efficiency
- proactive power upgrade planning

3. Literature Review

3.1. Telecommunication Network Energy Consumption

The rapid growth of mobile communication technologies has significantly increased the energy consumption of modern telecommunication networks. Over the past decade, the widespread deployment of Long-Term Evolution (LTE) and Fifth Generation (5G) technologies has dramatically increased both traffic throughput and infrastructure complexity within Radio Access Networks (RANs). Mobile network operators are continuously expanding network coverage, increasing spectral efficiency, deploying additional carriers, and supporting growing subscriber demand for high-speed mobile data services.

Among the various components of a telecommunication network, the Radio Access Network is widely recognized as one of the largest contributors to overall energy consumption. RAN infrastructure includes:

- Radio Remote Units (RRUs)
- Active Antenna Units (AAUs)
- Baseband Units (BBUs)
- transmission equipment
- cooling systems
- power conversion systems

3.2. Engineering-Based Power Estimation Approaches

Traditional telecommunication power planning primarily relies on engineering-rule-based estimation methods. These methods use predefined power consumption values derived from:

- vendor specifications and datasheets
- laboratory measurements
- operational experience
- engineering assumptions

In practical deployment scenarios, telecommunication engineers estimate site power demand by assigning nominal power values to individual hardware components such as:

- RRUs
- AAUs
- BBUs
- microwave radios
- cooling systems
- transmission equipment

The total site power is then calculated by aggregating the estimated contribution of each installed component. Engineering-based methods remain popular in industry because they are:

- relatively simple
- easy to implement
- computationally inexpensive
- operationally interpretable

These methods also align closely with existing infrastructure planning workflows used by passive infrastructure and power engineering teams.

Such calculations are commonly used during:

- LTE modernization projects
- 5G rollout programs
- rectifier sizing
- battery bank planning
- generator dimensioning
- air-conditioning system design

3.3. Machine Learning in Telecommunication Analytics

Machine learning techniques have become increasingly important within telecommunication analytics due to their ability to model complex nonlinear relationships using large operational datasets. Modern telecommunication networks generate vast quantities of data from:

- network management systems
- traffic monitoring platforms
- fault management systems
- performance counters
- power monitoring systems
- subscriber activity logs

These datasets provide valuable opportunities for applying machine learning techniques to improve operational decision-making.

4. Dataset and Preprocessing

4.1. Dataset Overview

This project utilized multiple operational datasets obtained from Cable & Wireless Seychelles (C&WS) in order to develop and evaluate telecommunication site power prediction frameworks. The datasets consisted of time-series operational data collected from 4G and 5G traffic utilization of cells.

The primary datasets used in this project included:

File / Data Source	Content
Site Database from Sey.xlsx	2G, 3G. 4G and 5G cell information along with GPS coordinates
4G Traffic.xlsx	Traffic generated by each 4G cell, maximum within a 15 minute timeslot
5G Traffic.xlsx	Traffic generated by each 5G cell, maximum within a 15 minute timeslot
site_power from Sey.xlsx	Power captured / logged from the Rectifier unit
LTE Master Raw Data.xlsx	Derived from Site Database from Sey.xlsx
5G Raw Data.xlsx	Derived from Site Database from Sey.xlsx

Table 4-1: Dataset Overview – Files, Source and Content

The time resolution of the dataset was particularly important because power consumption within mobile networks varies dynamically according to traffic load and subscriber activity. Using 15-minute granularity allowed the models to capture daily traffic variation, peak-hour behavior, low-utilization periods and technology-specific demand fluctuations.

The project primarily focused on predicting site-level power consumption using traffic behavior, hardware inventory information, engineering-rule calculations and machine learning models. The dataset also supported comparative analysis between engineering-rule prediction, cell-wise machine learning, site-wise machine learning and hybrid ML approaches.

4.2. Data Sources and Characteristics

The project combined multiple operational datasets originating from different functional areas of the telecommunication network environment. The Site Database dataset contained hardware inventory and infrastructure configuration information for all operational sites.

The dataset included parameters such as Site_ID, site name, RRU counts, AAU counts, baseband equipment information and board-level hardware configuration.

Key hardware-related columns included RRU_2G, RRU_3G, RRU_4G, AAU_5G, Boards_4G, Boards_5G, BBU5900, BBU3900 and BBU3910. The LTE and 5G traffic datasets contained high-resolution traffic telemetry collected from operational network systems including Site_ID, Cell_ID, trigger_ID, timestamp information and traffic_load_mbps.

The site power dataset contained measured site-level electrical power consumption values. These measurements represented the actual operational power usage observed at telecommunication sites and were used as the ground-truth target variable during machine learning model training and evaluation. The final prediction datasets generally followed the structure:

Column	Description	Column	Description
Site_ID	Unique telecommunication site identifier	trigger_ID	15-minute operational interval identifier
date	Operational date	error	Prediction residual
site_power	Actual measured site power	predicted power	Model output predictions
datetime	Combined timestamp	error_percentage	Relative prediction error

Table 4-2: Structure of the predicted datasets

4.3. Data Integration

One of the major technical tasks within the project involved integrating multiple operational datasets originating from different telecommunication systems. Since traffic, hardware, and power information were stored separately, a unified analytical dataset had to be constructed before model development. The integration process primarily relied on the Site_ID, trigger_ID and datetime.

Site_ID served as the primary infrastructure identifier representing individual telecommunication sites across all datasets. Trigger_ID represented the operational 15-minute measurement interval and was used to synchronize traffic and power measurements across technologies.

The LTE and 5G traffic datasets were first aggregated according to the requirements of each prediction methodology. For site-wise machine learning approaches

- traffic values were aggregated at site level
- all cell-level traffic belonging to the same site and timestamp were summed together.

For cell-wise machine learning approaches:

- traffic remained separated at cell level
- power prediction was performed individually for each cell before aggregation.

The engineering-rule framework also required separate preprocessing stages for 2G and 3G power estimation, LTE power estimation and 5G power estimation.

Technology-specific outputs were later merged together in order to generate total site-level predicted power. After aggregation and preprocessing, all datasets were merged into unified analytical tables using Pandas merge operations.

4.4. Data Cleaning

The first preprocessing step involved ensuring consistent column naming and datatype formatting across all datasets. Since some datasets originated from different operational systems, certain fields required standardization to maintain consistency during merging and aggregation.

Missing values represented another important preprocessing challenge. Some hardware configuration fields contained null or empty values due to incomplete inventory records, optional hardware configurations and unavailable engineering data. Missing numerical values were typically replaced using fillna(0) in order to maintain compatibility with aggregation and machine learning operations.

Several hardware-related columns were explicitly converted into numeric format using datatype conversion, coercion handling, null replacement. This was especially important for RRU counts, AAU counts, BBU counts and board-level configuration features.

Timestamp consistency was another critical preprocessing requirement. Since all datasets operated at 15-minute granularity, traffic and power measurements needed to be synchronized accurately. Datetime columns were aligned to ensure proper temporal matching, consistent aggregation, accurate model training.

Additionally, technology-specific traffic aggregation operations were performed separately for LTE traffic amd 5G traffic. These aggregated traffic features later became key inputs for site-wise ML prediction and hybrid ML models.

The preprocessing pipeline therefore ensured that all datasets became consistent, synchronized, numerically compatible and analytically suitable for downstream machine learning and engineering-rule processing.

4.5. Feature Engineering

Feature engineering represented one of the most important components of the project because telecommunication site power consumption depends on multiple interacting operational variables rather than a single independent factor.

Several categories of features were generated during preprocessing, including traffic-based features, hardware-based features, technology-specific features and aggregated infrastructure features.

Traffic-based feature engineering primarily focused on generating total LTE traffic, total 5G traffic, site-level traffic aggregation and cell-level traffic aggregation.

For site-wise ML approaches, all LTE cell traffic associated with a site was aggregated into total_4g_traffic and similarly, all 5G traffic was aggregated into total_5g_traffic. These aggregated traffic variables became primary predictive inputs for the site-wise ML models. For cell-wise approaches, traffic values remained associated with individual cells during prediction before final aggregation to site level.

Hardware-related feature engineering was performed using the Site Database dataset. Features included number of RRUs, number of AAUs, number of LTE boards, number of 5G boards and BBU hardware availability.

Additional derived features were also generated, including total_rru, total_5g_equipment, total_4g_equipment, total_bbu and legacy_equipment. These derived features improved the ability of machine learning models to capture infrastructure complexity and equipment composition.

Technology-specific power estimation features were also created during engineering-rule processing. These included power_2g, power_3g, power_4g, power_5g.

Finally, the engineering-rule framework generated site_total_predicted_power, by aggregating the predicted contribution of all technologies. Residual and error analysis features were also created, including error, error_percentage, absolute_error and residual_error.

4.6. Dataset Preparation for Different Models

Different preprocessing strategies were required for each of the four power prediction methodologies implemented in this project.

4.6.1. Engineering-Rule Dataset Preparation

The engineering-rule framework used hardware inventory information and telecom engineering assumptions to estimate technology-specific power consumption. Separate preprocessing pipelines were developed for 2G, 3G, LTE and 5G NR.

Each technology generated intermediate power outputs which were later aggregated to produce total site-level predicted power. The engineering-rule framework simulated practical telecom infrastructure planning processes commonly used by operational engineering teams.

4.6.2. Cell-Wise Machine Learning Dataset Preparation

The cell-wise ML framework processed traffic information at individual cell level. Machine learning models estimated power contribution separately for each cell before aggregating all predicted outputs to site level. This approach required cell-level preprocessing, traffic decomposition, cell aggregation and timestamp synchronization.

The final dataset produced Fully_Data_Driven_Cell_Wise_Predictions.xlsx and its corresponding summary outputs.

4.6.3. Site-Wise Machine Learning Dataset Preparation

The site-wise ML framework aggregated all traffic information at site level before performing prediction. The final dataset directly modeled the relationship between aggregated traffic and measured site power. The output dataset produced Fully_Data_Driven_Site_Wise_Predictions.xlsx.

4.6.4. Final Hybrid ML Dataset Preparation

The final ML framework combined aggregated traffic features, hardware inventory features and infrastructure characteristics. This produced a richer feature space capable of modeling traffic-dependent power behavior and hardware-dependent baseline power demand.

The final output dataset generated Final_Site_Power_Predictions.xlsx which achieved the best overall prediction accuracy among all evaluated methods.

5. Methodology

5.1. Overall System Architecture

This project implemented an end-to-end analytics and machine learning framework for predicting telecommunication site power consumption using operational traffic telemetry and hardware configuration data. The overall methodology combined:

- Engineering-Rule Calculations
- Data Preprocessing
- Feature Engineering
- Machine Learning Prediction
- Residual Analysis
- Comparative Evaluation

The system architecture followed a modular pipeline design consisting of

- Data Ingestion
- Preprocessing
- Cleaning
- Feature Engineering
- Technology-Specific Processing
- Machine Learning Prediction
- Performance Evaluation
- Residual And Risk Analysis

The overall workflow integrated multiple operational datasets including LTE traffic telemetry, 5G traffic telemetry, site hardware inventory and measured site power consumption.

The preprocessing layer was responsible for data synchronization, timestamp alignment, traffic aggregation, hardware feature preparation and dataset integration.

Following preprocessing, the project implemented four independent prediction methodologies:

- engineering-rule-based prediction
- fully data-driven cell-wise ML prediction
- fully data-driven site-wise ML prediction
- final hybrid machine learning framework

Each methodology generated predicted site power values which were later compared against actual measured site power data using multiple regression evaluation metrics.

The system architecture was designed to simulate a realistic telecom operational analytics pipeline suitable for infrastructure planning and decision support applications. The architecture also enabled comparative analysis between conventional telecom engineering approaches and modern machine learning approaches.

The overall system workflow can be summarized as follows.

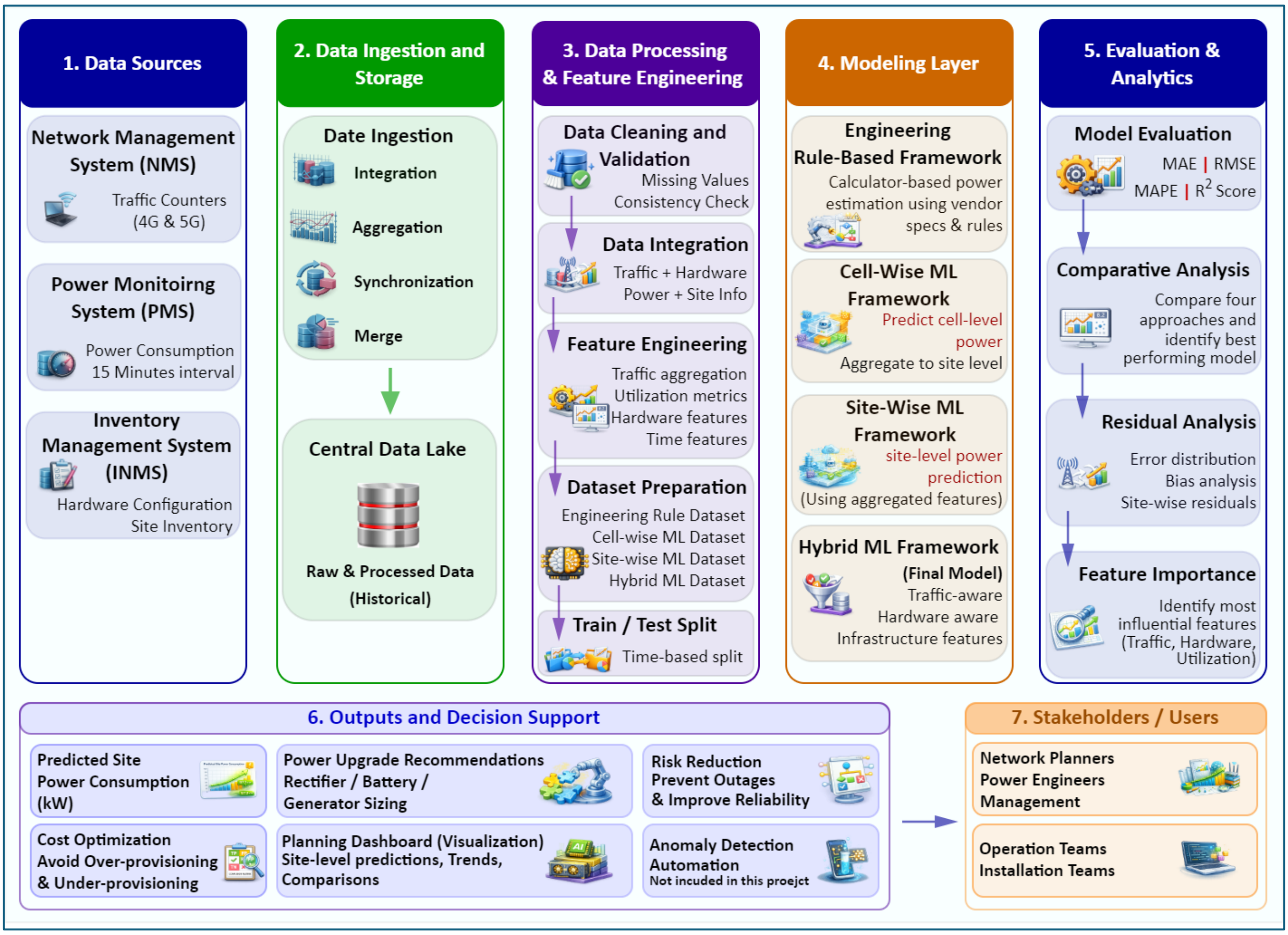

Figure 5-1: Overall System Architecture

5.2. Engineering Rule-Based Power Prediction Framework

The engineering-rule framework was developed to simulate traditional telecommunication power planning methodologies commonly used within operational engineering environments. This framework estimated site-level power consumption using predefined hardware power assumptions derived from engineering knowledge and vendor-based telecom planning practices.

The engineering-rule approach was implemented using four major components including 2G and 3G power estimation, LTE power estimation, 5G NR power estimation and total site power aggregation.

The methodology utilized hardware inventory information from the Site Database dataset, including RRU counts, AAU counts, BBU models, LTE boards and 5G boards.

For the 2G and 3G framework, static power values were assigned to radio units based on predefined engineering assumptions. Total 2G and 3G power consumption was estimated using RRU quantity, predefined power-per-unit values and site-level aggregation.

The LTE engineering framework utilized LTE traffic and hardware configuration information to estimate 4G site power. Similarly, the 5G engineering framework estimated NR power consumption using traffic load, radio power behavior and engineering-based scaling assumptions.

The specific technology outputs were generated as follows.

- power_2g3g
- power_4g
- power_5g

These values were aggregated together to generate the final site_total_predicted_power.

The engineering-rule framework simulated real-world telecom infrastructure planning activities such as:

- rectifier sizing
- battery bank planning
- power upgrade estimation
- hardware deployment assessment

One of the major advantages of this methodology was its strong operational interpretability. Since calculations were directly linked to physical hardware quantities and engineering assumptions, infrastructure planning teams could easily understand the prediction logic.

5.3. Fully Data-Driven Cell-Wise Machine Learning Framework

The first machine learning methodology implemented in this project was a fully data-driven cell-wise power prediction framework. This approach attempted to estimate power consumption at individual cell level before aggregating predictions to total site level.

The motivation behind this methodology was based on the assumption that telecommunication traffic originates at radio cell level. Therefore, modeling power consumption at finer analytical granularity could potentially improve prediction detail and capture localized traffic behavior more effectively.

The cell-wise framework utilized:

- LTE traffic data
- 5G traffic data
- cell-level traffic measurements
- operational timestamps
- site identifiers

Traffic values associated with individual radio cells were processed separately during model training and prediction. Machine learning models were trained to learn the relationship between cell traffic load and predicted cell power contribution.

After predicting cell-level power values, all predicted outputs belonging to the same site were aggregated together to estimate total site-level power consumption.

The framework primarily utilized Random Forest regression because of its ability to:

- model nonlinear relationships
- handle heterogeneous telecom datasets
- reduce overfitting
- support robust regression performance

The final output of the framework generated *Fully_Data_Driven_Cell_Wise_Predictions.xlsx* along with corresponding summary datasets and site-wise error analysis outputs.

Although the cell-wise methodology successfully demonstrated end-to-end machine learning implementation, several important limitations emerged during evaluation.

One of the major challenges involved the absence of true cell-level power measurements within the operational dataset. Since actual power measurements were available primarily at site level, the framework required indirect decomposition assumptions in order to estimate cell-level contribution.

Additionally, the aggregation process introduced cumulative prediction error. Small inaccuracies generated at individual cell level accumulated when aggregated to total site-level predictions. This error significantly affected overall prediction accuracy.

The cell-wise framework therefore demonstrated higher residual variability, larger aggregation noise and weaker overall predictive performance compared to site-wise approaches.

Despite these limitations, the framework provided valuable analytical insight into decomposition-based telecom prediction, granularity-related modeling challenges and aggregation error propagation within multi-cell telecommunication environments.

5.4. Fully Data-Driven Site-Wise Machine Learning Framework

The second machine learning methodology implemented in this project was a fully data-driven site-wise power prediction framework. Unlike the cell-wise approach, this methodology aggregated all traffic information at site level before performing prediction.

The primary objective of this methodology was to directly model the relationship between aggregated site traffic and measured site power consumption.

The site-wise framework utilized

- aggregated LTE traffic
- aggregated 5G traffic
- operational timestamps
- site-level power measurements

All traffic values associated with a site were aggregated according to Site_ID, trigger_ID, and datetime.

The resulting features included total_4g_traffic and total_5g_traffic.

The machine learning model then directly predicted site-level power consumption without requiring intermediate cell-level decomposition.

This approach significantly simplified,

- preprocessing
- feature engineering
- synchronization
- prediction workflows

More importantly, direct site-level learning eliminated the aggregation error propagation observed within the cell-wise framework.

The Random Forest regression model again served as the primary prediction algorithm due to its strong capability to model nonlinear traffic-power relationships, heterogeneous operational features and dynamic telecom behavior.

The site-wise methodology generated *Fully_Data_Driven_Site_Wise_Predictions.xlsx* and corresponding residual analysis outputs.

The site-wise framework demonstrated a major improvement in predictive performance compared to the cell-wise methodology. The direct learning approach reduced decomposition complexity, synthetic label dependency and cumulative aggregation noise.

This finding became one of the most important technical observations within the project. The results demonstrated that direct site-level learning is significantly more effective than synthetic cell-level decomposition for telecom power prediction.

The methodology also aligned more closely with practical telecom operational environments because actual site power measurements are naturally collected at site level rather than individual cell level.

5.5. Final Hybrid Machine Learning Framework

The final and best-performing methodology implemented in this project was a hybrid machine learning framework combining traffic information, hardware inventory features and infrastructure characteristics.

This framework extended the site-wise ML methodology by integrating telecom hardware configuration parameters directly into the machine learning feature space.

The model utilized aggregated traffic features such as total_4g_traffic and total_5g_traffic together with hardware-related features including:

- RRU_2G
- RRU_3G
- RRU_4G
- AAU_5G
- Boards_4G
- Boards_5G
- BBU5900
- BBU3900
- BBU3910

Additional derived infrastructure features were also generated, including:

- total_rru
- total_5g_equipment
- total_4g_equipment
- total_bbu
- legacy_equipment

The final model therefore combined traffic-driven operational behavior, hardware-dependent baseline power demand and infrastructure complexity indicators.

Unlike the engineering-rule framework, the hybrid ML model did not rely on manually assigned wattage values, static vendor assumptions and predefined telecom equations.

Instead, the Random Forest algorithm learned complex nonlinear relationships directly from historical operational data.

The final hybrid framework generated *Final_Site_Power_Predictions.xlsx* which achieved the strongest overall performance among all evaluated methodologies.

The framework demonstrated:

- lowest prediction error
- highest R^2 value
- strongest generalization capability
- best operational suitability

This methodology represented the most scalable and adaptive solution developed within the project.

5.6. Machine Learning Algorithm Selection

Random Forest regression was selected as the primary machine learning algorithm throughout the project due to several important technical advantages.

Random Forest is an ensemble learning technique that combines multiple decision trees in order to improve predictive accuracy and reduce overfitting. The algorithm operates by generating multiple randomized trees during training and averaging their outputs during prediction.

The algorithm was particularly suitable for telecommunication power prediction because telecom operational datasets contain:

- nonlinear relationships
- mixed feature types
- heterogeneous operational behavior
- dynamic traffic interactions

Random Forest also performs effectively when:

- feature interactions are complex
- exact mathematical relationships are unknown
- operational variability is high.

Additional advantages of Random Forest are,

- robustness against overfitting
- ability to handle missing values
- strong regression performance
- minimal normalization requirements
- feature importance analysis capability

The algorithm therefore provided a practical balance between predictive accuracy, operational interpretability and implementation simplicity.

During early experimentation (not published along with the Final Report), additional models such as XGBoost, Prophet, NeuralProphet and LSTM were also investigated. However, Random Forest demonstrated the most stable and operationally suitable performance within the available dataset constraints.

5.7. Model Evaluation Strategy

All prediction frameworks were evaluated using multiple regression performance metrics in order to assess prediction accuracy, residual behavior and operational suitability.

The primary evaluation metrics included,

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- coefficient of determination (R^2).

MAE measured the average absolute difference between actual measured site power and predicted site power. This metric provided direct operational interpretation in terms of wattage deviation.

RMSE measured the square-root of mean squared prediction error. Since RMSE penalizes larger errors more heavily, it provides additional insight into prediction stability, sensitivity to large residuals and operational risk behavior.

MAPE measured relative percentage errors and was particularly important for telecom infrastructure planning because rectifier sizing, battery planning and upgrade estimation depend heavily on relative prediction accuracy rather than only absolute wattage deviation.

The R^2 metric measured how effectively each model explained variability within the observed site power data.

All models were additionally evaluated using,

- scatter plots
- residual analysis
- site-wise comparison
- error histograms
- feature importance analysis

This combination of quantitative and visual evaluation techniques provided comprehensive assessment of model behavior.

5.8. Residual Error Analysis

Residual analysis formed an important component of the project because prediction errors within telecommunication power planning can create direct operational consequences.

Residual error = ACTUAL SITE POWER − PREDICTED SITE POWER

Positive residual values indicated underestimation where actual site power exceeded predicted power. Negative residual values indicated overestimation where predicted values exceeded actual operational demand. Underestimation represents a particularly important operational risk because insufficient power provisioning may lead to,

- rectifier overload
- battery stress
- site outages
- infrastructure instability

Overestimation, although operationally safer, may increase unnecessary capital expenditure, oversized infrastructure deployment and inefficient utilization.

Residual analysis was therefore performed using:

- site-wise error plots
- absolute error graphs
- residual histograms
- maximum error summaries

This analysis enabled identification of:

- high-risk sites
- unstable prediction regions
- operational outliers
- infrastructure planning concerns

5.9. Comparative Framework Design

One of the major contributions of this project was the development of a unified comparative framework for evaluating multiple telecommunication power prediction methodologies within the same operational dataset environment.

The framework compared,

- engineering-rule prediction
- cell-wise machine learning prediction
- site-wise machine learning prediction
- hybrid traffic-and-hardware-aware ML prediction.

This comparative design enabled detailed evaluation of:

- predictive accuracy
- operational interpretability
- residual behavior and Scalability
- infrastructure planning suitability

The engineering-rule framework provided strong operational transparency, telecom engineering interpretability and practical infrastructure planning alignment. However, the ML frameworks provided adaptive learning capability, nonlinear relationship modeling and improved prediction accuracy.

Among the ML approaches, the site-wise methodologies significantly outperformed the cell-wise framework due to reduced decomposition complexity and lower aggregation error propagation.

The final hybrid ML framework achieved the best overall performance by combining traffic-aware learning, infrastructure-aware feature engineering and operational telecom context.

The comparative methodology therefore demonstrated that data-driven site-level machine learning frameworks provide superior predictive performance for telecommunication power upgrade planning compared to traditional decomposition-based or purely engineering-rule approaches.

6. Results and Discussion

6.1. Overview of Experimental Evaluation

This chapter presents the experimental results and comparative evaluation of the four telecommunication site power prediction methodologies implemented in this project. The primary objective of the evaluation was to assess,

- prediction accuracy
- operational suitability
- residual behavior
- scalability
- infrastructure planning applicability

across multiple engineering-rule and machine learning approaches.

The evaluated methodologies included,

- Engineering-rule-based prediction framework
- Fully data-driven cell-wise ML framework
- Fully data-driven site-wise ML framework
- Final hybrid machine learning framework

All prediction outputs were compared against actual measured site-level power consumption obtained from operational telecommunication infrastructure datasets.

The evaluation process combined,

- quantitative regression metrics
- graphical analysis
- residual analysis
- feature importance evaluation
- site-wise comparison
- operational interpretation

The analysis focused not only on statistical accuracy but also on practical infrastructure planning implications such as power upgrade sizing, rectifier loading risk, operational stability and infrastructure investment optimization.

The project generated prediction outputs for 75 telecommunication sites across 50,400 operational records covering 7 consecutive days, 15-minute operational intervals, multi-RAT network environments.

The final results demonstrated significant differences in performance between engineering-rule approaches, cell-wise ML approaches and site-wise ML approaches.

One of the most important findings of the project was that direct site-level machine learning significantly outperformed cell-wise decomposition-based prediction methodologies.

6.2. Evaluation Metrics

The prediction frameworks were evaluated using four primary regression metrics,

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- coefficient of determination (R^2)

These metrics were selected because they provide complementary perspectives regarding prediction quality and operational suitability.

6.2.1. Mean Absolute Error (MAE)

MAE measured the average absolute deviation between actual site power and predicted site power.

This metric provided direct operational interpretation in terms of average wattage error. Lower MAE values indicated more accurate prediction performance. Within telecommunication infrastructure planning, MAE is important because large wattage deviations may result in,

- incorrect rectifier sizing
- battery under-dimensioning
- power overload conditions
- unnecessary infrastructure upgrades

6.2.2. Root Mean Squared Error (RMSE)

RMSE measured the square-root of mean squared prediction error. Since RMSE penalizes larger residuals more heavily than MAE, it provided insight into:

- prediction stability
- sensitivity to large operational deviations
- worst-case infrastructure risk behavior.

High RMSE values generally indicate unstable prediction behavior or large residual outliers.

6.2.3. Mean Absolute Percentage Error (MAPE)

MAPE measured the relative percentage difference between actual power and predicted power.

MAPE was particularly important for this project because telecommunication power planning decisions are often based on relative estimation margins rather than only absolute wattage values.

For example:

- a 5% prediction error may be operationally acceptable
- a 25% prediction error may lead to major infrastructure planning issues.

MAPE therefore provided strong operational interpretation regarding upgrade planning accuracy, infrastructure sizing reliability and deployment confidence.

6.2.4. Coefficient of Determination (R^2)

The R^2 metric measured how effectively each model explained variability within the observed site power dataset. Higher R^2 values indicates stronger predictive capability, improved model fitting, better representation of operational behavior

R^2 values close to 1.0 indicates that the prediction model successfully captured most variability within the measured site power data.

6.3. Comparative Performance Summary

The table below summarizes the overall performance achieved by the four evaluated methodologies.

Methodology	MAE	RMSE	MAPE	R^2
Cell-Wise ML	459.09	640.08	11.88%	0.9034
Site-Wise ML	154.95	217.58	3.80%	0.9888
Engineering Rule	175.14	251.00	4.15%	0.9851
Final Hybrid ML	152.10	213.10	3.74%	0.9893

Table 6-1: Comparative Performance Summary

The results clearly demonstrate substantial performance differences between the evaluated methodologies.

The fully data-driven cell-wise ML framework produced the weakest performance among all approaches, achieving, highest MAE, highest RMSE, highest MAPE and lowest R^2.

In contrast, the final hybrid machine learning framework achieved the strongest overall performance with lowest MAE, lowest RMSE, lowest MAPE and highest R^2.

The engineering-rule framework also demonstrated surprisingly strong predictive capability despite relying primarily on predefined telecom engineering assumptions. This finding indicates that telecom engineering knowledge remains highly valuable for infrastructure planning applications even within modern machine learning environments.

6.4. Engineering Rule-Based Prediction Results

The engineering-rule framework achieved:

- MAE ≈ 175.14
- RMSE ≈ 251.00
- MAPE ≈ 4.15%
- $R^2 \approx 0.9851$

These results demonstrated that engineering-rule methodologies remain operationally effective for telecommunication infrastructure planning when appropriate telecom hardware assumptions are used.

The relatively low MAPE value indicated that static engineering assumptions were capable of approximating real operational behavior with reasonable accuracy.

This strong performance can be attributed to the fact that telecommunication site power consumption is heavily influenced by:

- installed hardware inventory
- transmission equipment quantity
- radio unit configuration
- technology composition

Since the engineering-rule framework directly modeled these physical infrastructure characteristics, the methodology successfully captured the baseline behavior of site power demand.

Another important advantage of the engineering-rule approach was its strong interpretability. Infrastructure planning engineers could directly understand,

- how power was calculated
- which hardware contributed most
- how individual technologies affected total site demand.

However, the engineering-rule framework also demonstrated several limitations. The methodology relied heavily on static vendor assumptions, predefined power values and simplified utilization behavior.

As a result, the framework struggled to fully capture:

- dynamic traffic variation
- nonlinear operational behavior
- temporal utilization fluctuations
- complex traffic-power relationships

Residual analysis also showed that certain high-traffic sites experienced larger underestimation errors due to operational conditions not represented within static engineering assumptions.

Nevertheless, the engineering-rule model provided an important operational baseline against which machine learning approaches could be compared.

6.5. Fully Data-Driven Cell-Wise ML Results

The fully data-driven cell-wise ML framework achieved:

- MAE ≈ 459.09
- RMSE ≈ 640.08
- MAPE ≈ 11.88%
- $R^2 \approx 0.9034$

These results represented the weakest performance among all evaluated methodologies.

Although the framework successfully demonstrated end-to-end machine learning implementation using cell-level traffic data, several important technical limitations reduced prediction accuracy.

The largest challenge involved the absence of directly measured cell-level power data. Since operational power measurements were available primarily at site level, the framework required indirect decomposition assumptions to estimate individual cell contribution.

As a result, the machine learning model learned from partially synthetic or approximated relationships rather than directly measured cell-level power behavior.

Additionally, prediction errors accumulated during aggregation. Since multiple-cell-level predictions were summed together to estimate total site-level power, small inaccuracies at individual cell level propagated into larger site-level residuals.

This noise aggregation significantly increased residual variability, instability and overall prediction error.

The cell-wise framework also required more complex preprocessing operations involving,

- cell synchronization
- timestamp matching
- traffic decomposition
- site-level aggregation

Despite its weaker performance, the cell-wise methodology provided several important research insights.

First, the framework demonstrated that increased analytical granularity does not necessarily improve predictive accuracy within telecommunication power prediction environments.

Second, the results highlighted the importance of label quality, aggregation strategy and operational measurement availability within machine learning system design.

Finally, the cell-wise framework established an important comparative baseline demonstrating the limitations of decomposition-based telecom power prediction.

6.6. Fully Data-Driven Site-Wise ML Results

The fully data-driven site-wise ML framework achieved,

- MAE ≈ 154.95
- RMSE ≈ 217.58
- MAPE ≈ 3.80%
- $R^2 \approx 0.9888$

This represented a dramatic improvement compared to the cell-wise ML methodology.

The major improvement was primarily caused by direct site-level learning. Instead of predicting individual cell power and later aggregating outputs, the site-wise framework directly learned the relationship between aggregated site traffic and measured site power.

This eliminated decomposition-related noise, aggregation error propagation and synthetic label dependency.

The site-wise framework also simplified preprocessing, synchronization, feature engineering and prediction workflows.

Direct site-level prediction aligned naturally with the available operational measurements because actual site power was already monitored at site level within the telecommunication infrastructure.

Another major advantage of the site-wise methodology involved improved operational scalability. Since the framework required fewer intermediate calculations and reduced preprocessing complexity, it became more suitable for,

- nationwide deployment
- large-scale infrastructure analysis
- operational automation.

The site-wise results demonstrated that direct site-level learning is significantly more effective than cell-level decomposition for telecom site power prediction. This finding became one of the strongest technical conclusions of the project.

6.7. Final Hybrid ML Framework Results

The final hybrid machine learning framework achieved the strongest overall performance among all evaluated methodologies,

- MAE ≈ 152.10
- RMSE ≈ 213.10
- MAPE ≈ 3.74%
- $R^2 \approx 0.9893$

The hybrid framework extended the site-wise ML approach by integrating

- traffic information
- telecom hardware configuration
- infrastructure-related features

This richer feature space enabled the Random Forest model to simultaneously learn:

- dynamic traffic behavior
- hardware-dependent baseline demand
- infrastructure complexity effects.

The inclusion of hardware inventory features significantly improved the model's ability to capture,

- persistent baseline power consumption
- technology-specific operational differences
- infrastructure scaling behavior

Unlike the engineering-rule methodology, the hybrid ML framework did not rely on manually assigned telecom power equations. Instead, the model learned operational relationships directly from historical data.

The final hybrid model also demonstrated strong generalization capability, lowest residual variability and highest predictive stability. Scatter plot analysis showed strong alignment between actual site power and predicted site power with most observations clustered closely around the ideal prediction line.

Residual analysis further demonstrated that the hybrid model produced fewer large outliers, lower site-wise maximum error, improved operational consistency.

These results indicate that combining traffic-aware learning, infrastructure-aware features and site-level prediction provides the most effective framework for telecommunication power upgrade planning.

6.8. Residual and Error Analysis

Residual analysis provided important operational insight into model behavior and infrastructure planning risk. RESIDUAL ERROR is considered as ACTUAL SITE POWER − PREDICTED SITE POWER.

Positive residuals indicated underestimation conditions where actual operational demand exceeded predicted values. Negative residuals indicated overestimation conditions where predicted power exceeded measured site demand.

Site-wise residual plots revealed that most sites remained within acceptable prediction margins and only a limited number of sites produced large residual outliers. The engineering-rule model generally exhibited moderate residual spread and occasional underestimation at high-traffic sites.

The cell-wise ML framework showed highest residual variability, largest site-wise errors and strongest aggregation instability. The site-wise and final hybrid ML frameworks demonstrated significantly tighter residual distribution, improved prediction stability and reduced operational risk. Absolute error analysis also identified a small number of high-risk sites requiring additional investigation.

6.9. Feature Importance Analysis

Feature importance analysis was performed using the Random Forest framework in order to identify the most influential variables affecting telecommunication site power prediction. The analysis demonstrated that aggregated traffic features represented some of the strongest predictive variables within the dataset.

The most influential features included total_4g_traffic, total_5g_traffic, AAU and RRU counts and total radio equipment quantity. This result confirms both traffic demand and installed hardware inventory play critical roles in determining site-level power consumption. Hardware-related features contributed strongly to baseline power demand, infrastructure scaling behavior and technology-specific operational characteristics.

Traffic-related features primarily influenced dynamic utilization behavior, temporal operational variation and real-time power fluctuation. The feature importance results therefore validated the design of the final hybrid ML framework which combined traffic-aware learning and hardware-aware feature engineering.

6.10. Discussion of Key Findings

Engineering-rule methodologies remain surprisingly effective for telecommunication infrastructure planning. Despite relying on static engineering assumptions, the engineering-rule framework achieved MAPE ≈ 4.15%, which represents relatively strong operational performance.

The results demonstrated that cell-level decomposition does not necessarily improve telecommunication power prediction accuracy. The cell-wise framework produced significantly weaker results due to synthetic decomposition assumptions, aggregation error propagation and lack of direct cell-level measurements.

The project demonstrated that direct site-level learning provides major advantages for telecom power prediction. The site-wise ML methodology significantly outperformed the cell-wise framework while simultaneously simplifying preprocessing and operational scalability.

The final hybrid ML framework demonstrated that combining traffic features, hardware inventory and infrastructure-aware feature engineering produces the strongest predictive capability for telecom infrastructure planning applications. The final model achieved MAPE ≈ 3.74% and $R^2 \approx 0.9893$ demonstrating excellent operational suitability for supporting power upgrade planning, infrastructure risk assessment, rectifier sizing and deployment of decision support.

The graphs below elaborate measured and predicted power by the 4 frameworks used in this project.

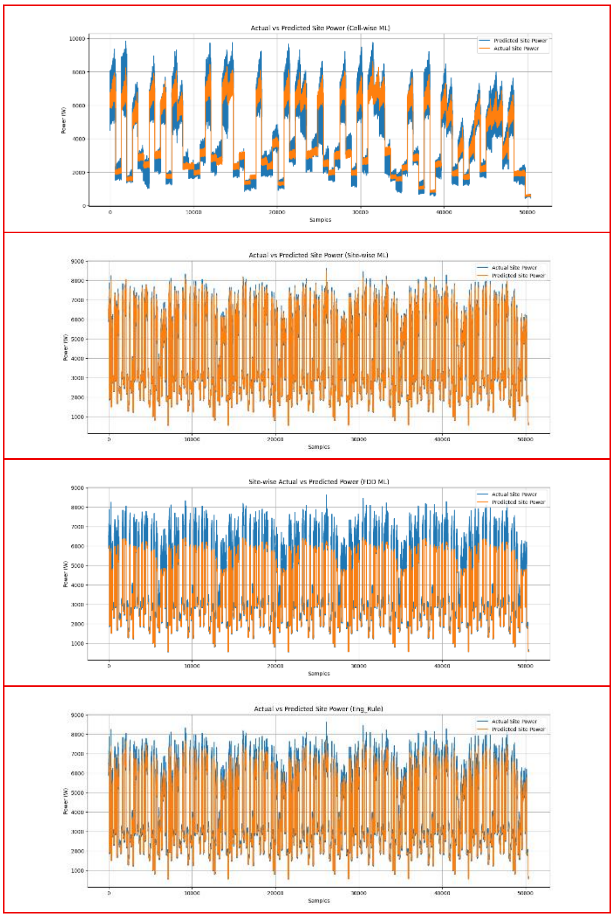

Figure 6-1: Actual Vs Predicted Power

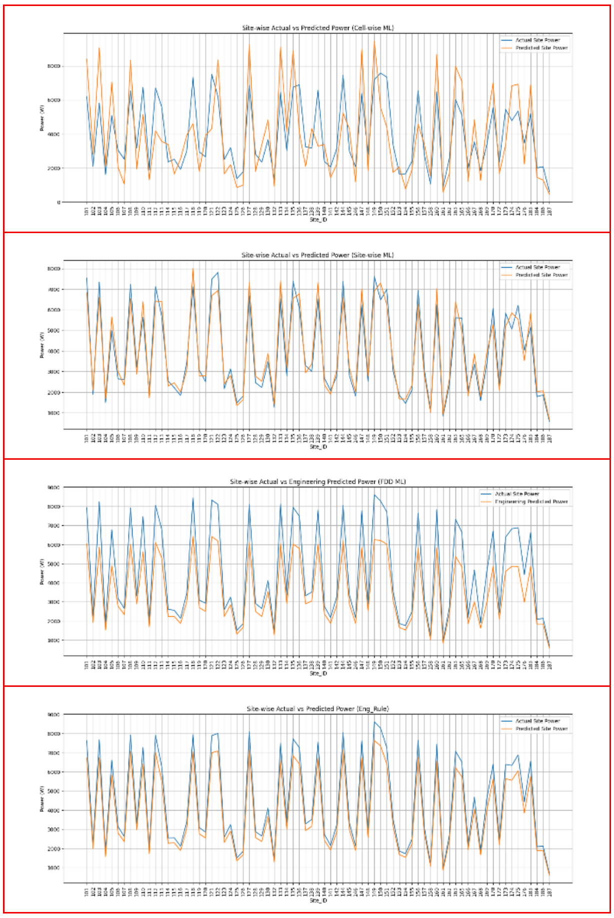

Figure 6-2: Actual Vs Predicted Power

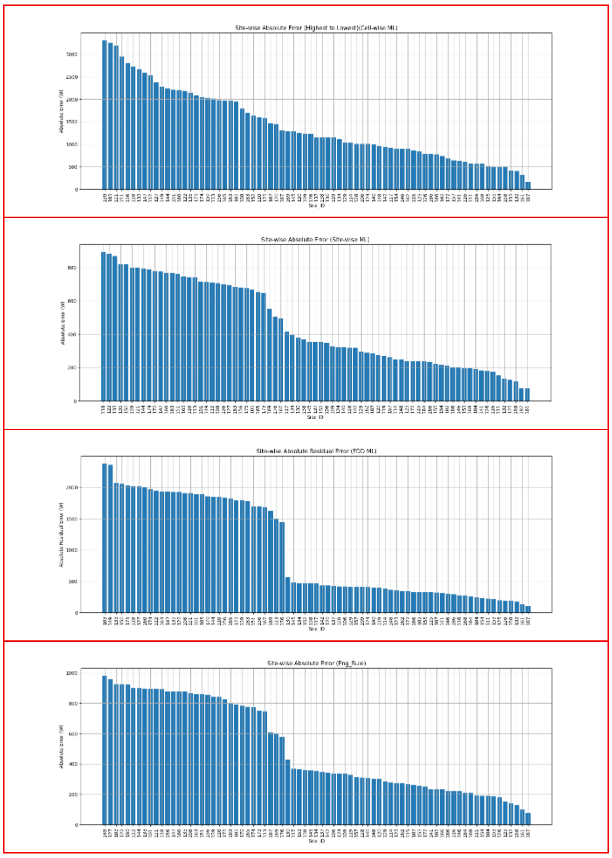

Figure 6-3: Absolute Error of Actual and Predicted power

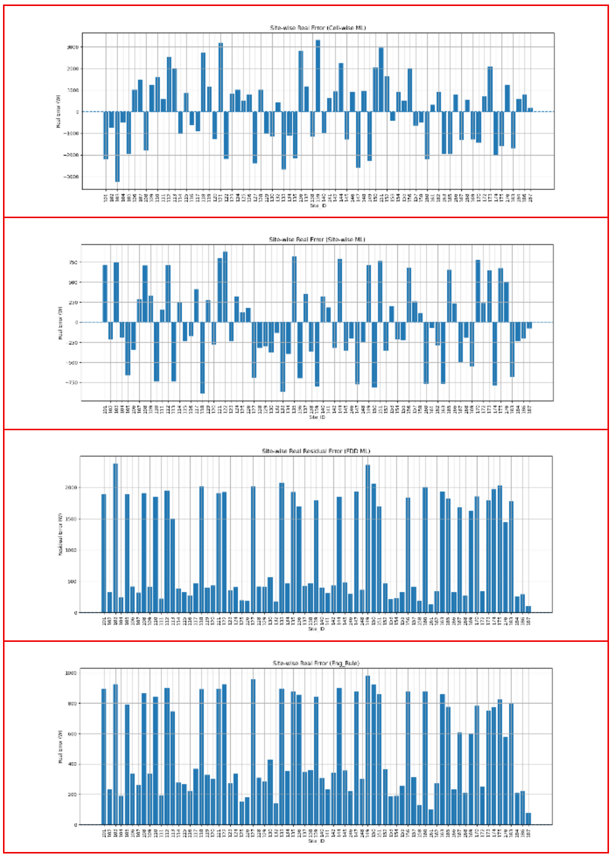

Figure 6-4: Real Error in Actual and Predicted power

The graphs below demonstrate the Scatterplots of Actual Vs Predicted power and Prediction Error Distribution of the 4 frameworks used in this project.

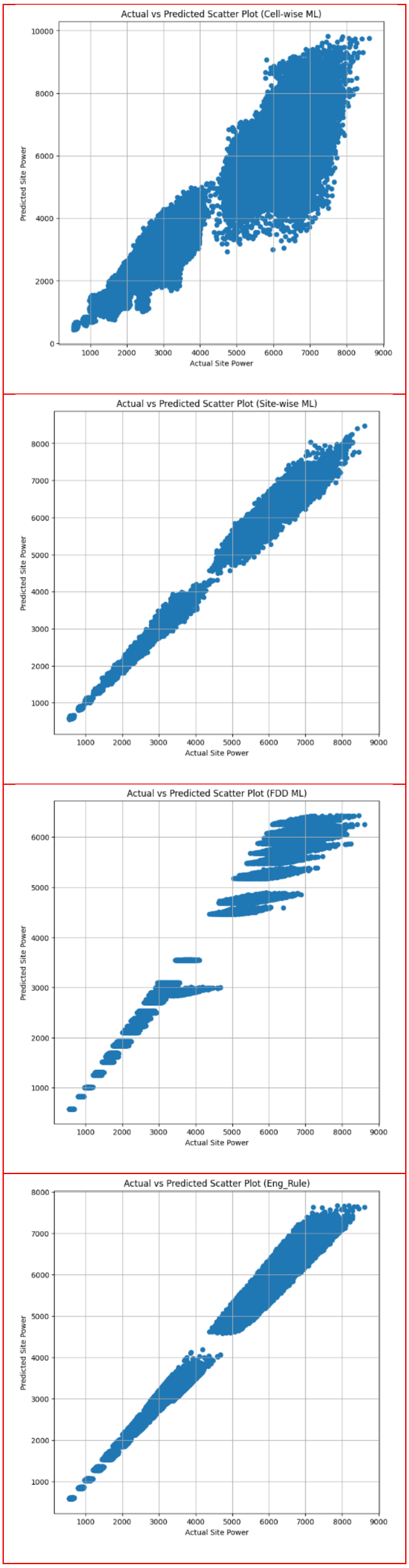

Figure 6-5: Actual vs Predicted Scatter Plot

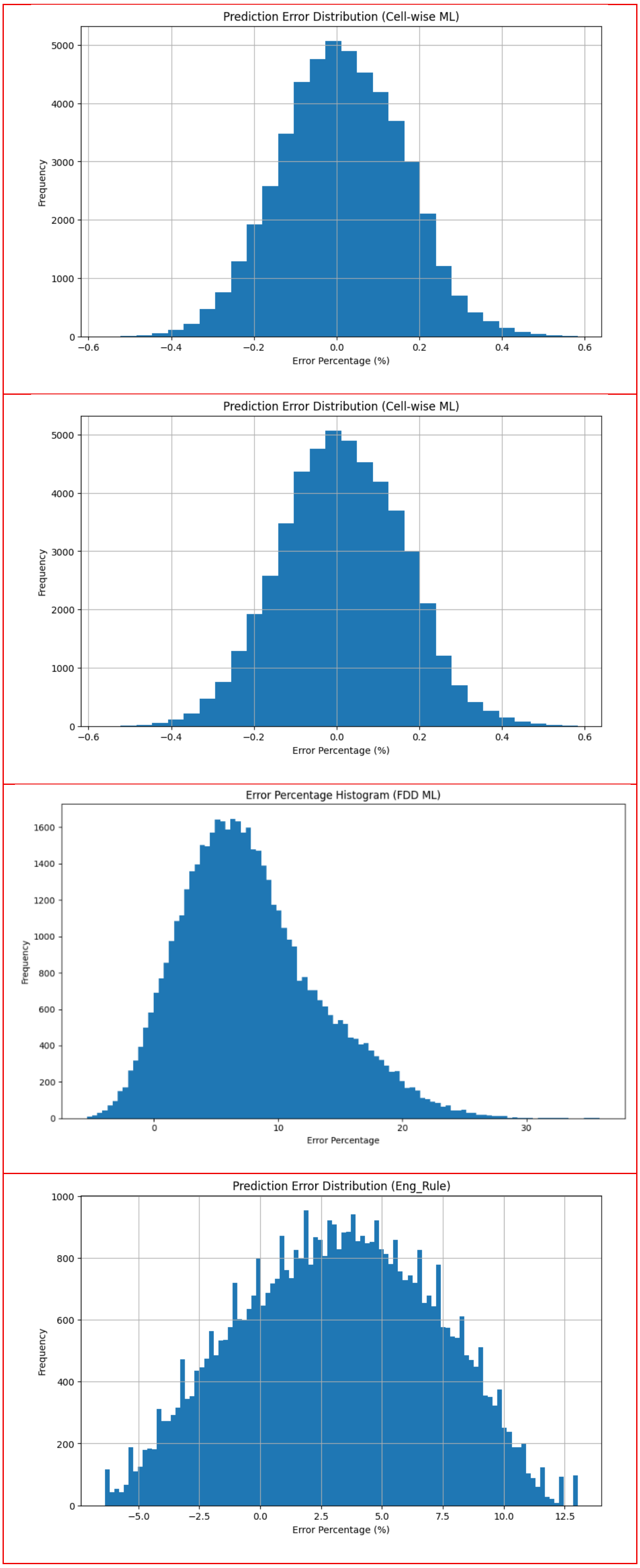

Figure 6-6: Prediction Error Distribution

7. Business Impact, Limitations, and Future Improvements

7.1. Business and Operational Impact

Accurate prediction of telecommunication site power consumption provides significant operational and financial value for mobile network operators. Modern Radio Access Networks (RANs) are continuously expanding due to increasing customer demand for mobile broadband services, video streaming, cloud applications, enterprise connectivity, Internet of Things (IoT) services, etc.

As operators deploy additional LTE and 5G infrastructure, site-level electrical demand increases substantially. Consequently, power system planning has become a critical component of telecommunication infrastructure modernization programs.

The framework developed in this project can directly support passive infrastructure planning teams responsible for rectifier sizing, battery bank planning, generator dimensioning, power upgrade approval, infrastructure capacity analysis, etc.

By accurately predicting site-level power demand, the proposed system can help operators avoid both under-provisioning and over-provisioning of telecommunication power systems.

One of the most important operational benefits of the proposed framework is reduction of outage risk. When site power systems are undersized, excessive electrical demand may overload rectifiers and battery systems, potentially causing voltage instability, thermal stress, battery degradation and complete site shutdown.

Since modern cellular sites commonly support 2G, 3G, LTE, 5G, microwave transmission systems, etc. a single site power failure may affect multiple communication services simultaneously. The proposed prediction framework enables infrastructure planning teams to proactively identify sites requiring rectifier upgrades, battery and power system expansion before operational overload conditions occur.

The framework also provides important financial benefits. Telecommunication power infrastructure components such as rectifiers, battery banks, generators, cooling systems, etc. represent major capital expenditure items during network modernization projects.

Traditional engineering-rule methods often apply highly conservative safety margins to avoid overload risk. While operationally safe, this may lead to oversized infrastructure deployment, inefficient resource utilization and unnecessary CAPEX expenditure.

The machine learning frameworks developed in this project provide more accurate prediction capability, enabling operators to optimize infrastructure investment decisions more effectively.

Another major operational advantage involves deployment acceleration. In many practical situations, operators purchase expensive telecom hardware such as RRUs, AAUs, baseband units and transmission equipment before corresponding power upgrades are completed. As a result, newly purchased hardware may remain unused until supporting electrical infrastructure becomes available.

By improving prediction accuracy and upgrading planning efficiency, the proposed framework can help operators reduce deployment delays, accelerate service activation and improve return on infrastructure investment.

The framework is also particularly valuable for geographically challenging deployment environments such as remote islands, mountainous regions, non-motorable sites, etc.

In such environments, emergency maintenance operations become operationally expensive and logistically difficult. Predictive infrastructure planning can therefore reduce emergency site visits, operational downtime, transportation cost, restoration delays etc.

From an organizational perspective, the proposed framework also supports movement toward data-driven infrastructure management, AI-assisted planning and intelligent operational analytics.

The framework demonstrates how existing operational datasets can be transformed into practical engineering decision-support systems within telecommunication environments.

7.2. Practical Deployment Considerations

Although the project was implemented as an academic capstone study, the proposed framework was designed with practical operational deployment considerations in mind.

The overall system architecture can be integrated into existing telecommunication operational environments using network management systems, power monitoring systems, traffic analytics platforms and centralized data lakes.

Since the framework primarily utilizes structured operational datasets already available within telecom operators, large-scale deployment would not require major additional infrastructure investment.

The prediction pipeline could operate periodically using hourly execution, daily execution and near-real-time operational updates. For example, operators could automatically generate daily power upgrade risk reports, high-risk site alerts, infrastructure utilization summaries and capacity forecasting outputs.

The framework could also be integrated into passive infrastructure planning workflows, modernization approval processes, upgrading budgeting systems and engineering dashboards.

Machine learning model retraining could be performed periodically using newly collected operational data. Continuous retraining would allow the framework to adapt automatically to changing traffic behavior, evolving hardware deployment, new technology introduction and network growth patterns.

One important operational advantage of the final hybrid ML framework is its scalability. Since prediction is performed directly at site level, the framework can efficiently support nationwide deployment, large multi-site analysis and centralized infrastructure planning.

The framework also maintains relatively strong operational interpretability compared to more complex deep learning systems. Infrastructure engineers can understand how traffic demand, hardware inventory, and infrastructure composition affect prediction behavior. This improves organizational acceptance and increases practical deployment feasibility.

7.3. Project Limitations

The most significant limitation involved dataset duration. The operational dataset covered only 7 consecutive days of measurements. This limited the ability to analyze seasonal traffic variation, long-term behavioral trends, special-event traffic patterns and long-duration infrastructure behavior.

A longer historical dataset covering several months or years would likely improve model generalization, seasonal learning capability and long-term prediction stability.

Another important limitation involved incomplete traffic telemetry for legacy technologies. Detailed operational traffic data was primarily available for LTE & 5G NR while 2G & 3G prediction relied more heavily on engineering-rule assumptions. Consequently, the project could not fully implement traffic-aware ML prediction for all technologies simultaneously.

The project also lacked several potentially valuable operational variables such as environmental temperature, cooling system behavior, battery health information, generator activity and commercial power failure history. These variables may significantly influence total site power consumption under certain operational conditions.

Another limitation involved the absence of true cell-level power measurements. Since operational power monitoring was performed primarily at site level, the cell-wise ML framework required indirect decomposition assumptions. This contributed significantly to aggregation error, residual instability and weaker predictive performance.

Operational confidentiality restrictions also limit the ability to disclose certain vendor-specific parameters, infrastructure details, planning thresholds and network configuration information.

Finally, the project focused primarily on prediction accuracy, comparative evaluation rather than real-time deployment or live operational integration.

Despite these limitations, the project successfully demonstrated the feasibility and operational value of data-driven telecom power prediction frameworks.

7.4. Future Improvements

Several opportunities exist for extending and improving the proposed framework in future work.

One major improvement would involve utilizing longer historical datasets covering multiple months, seasonal variation, holiday periods, special traffic events and long-term infrastructure evolution.

Longer datasets would improve model robustness, temporal learning capability and forecasting stability.

Future studies could also integrate additional operational variables such as:

- environmental temperature
- humidity
- cooling system performance
- battery aging behavior
- generator utilization
- commercial power availability.

These variables would likely improve prediction accuracy further, particularly for sites operating under challenging environmental conditions.

Another important future improvement involves implementation of advanced deep learning models such as Long Short-Term Memory (LSTM), Transformer architecture, Temporal Convolutional Networks and attention-based forecasting systems.

Such models may better capture temporal traffic dynamics, sequential operational behavior and long-term dependency patterns.

Future research could also explore anomaly detection, predictive maintenance, infrastructure risk scoring, unsupervised clustering and AI-assisted capacity planning.

Anomaly detection systems could automatically identify abnormal power behavior, inefficient sites, faulty infrastructure and unexpected traffic-power relationships.

The framework could additionally be extended into a real-time operational decision-support platform integrated with:

- Network Operations Centers (NOCs)
- infrastructure planning systems
- monitoring dashboards
- alarm management systems.

Future implementations may also investigate reinforcement learning approaches capable of dynamic infrastructure optimization, intelligent power allocation and adaptive energy management.

Finally, future research could evaluate transferability of the framework to other operators, larger national networks, multi-country deployments and different telecommunication vendor environments.

Such studies would help validate the scalability and generalizability of the proposed methodologies across broader telecom operational contexts.

8. Conclusion

8.1. Project Summary

The rapid expansion of modern telecommunication networks has significantly increased the complexity of infrastructure planning and energy management within Radio Access Networks (RANs). Mobile network operators continuously deploy LTE infrastructure, 5G equipment, additional radio capacity and advanced transmission systems in order to satisfy increasing subscriber demand for high-speed communication services.

These infrastructure upgrades substantially increase site-level electrical demand and create major challenges for rectifier sizing, battery planning, generator capacity analysis and infrastructure upgrade approval. This project is focused on developing and evaluating multiple methodologies for predicting telecommunication site power consumption using operational traffic telemetry, hardware inventory information, engineering-rule calculations and machine learning techniques.

The project utilized real operational datasets obtained from Cable & Wireless Seychelles (C&WS), including LTE & 5G traffic data, site hardware configuration and measured site power consumption. The dataset represented 75 operational telecommunication sites, 15-minute measurement intervals, 7 consecutive days of operational activity and multi-RAT network environments.

Four independent power prediction methodologies (frameworks) were implemented and evaluated;

- Engineering-rule-based prediction
- Fully data-driven site-wise ML
- Fully data-driven cell-wise ML
- Final hybrid machine learning

The project also implemented preprocessing pipelines, traffic aggregation frameworks, feature engineering workflows, residual analysis systems and comparative evaluation methodologies.

Comprehensive performance evaluation was performed using:

- MAE, RMSE, MAPE, R^2
- site-wise error analysis
- scatter plots, residual analysis
- feature importance analysis

The project successfully demonstrated the feasibility of using machine learning techniques for telecommunication infrastructure planning and power prediction applications.

8.2. Key Findings

Description	Estimators
Engineering-rule methodologies remain operationally valuable for telecommunication infrastructure planning. This demonstrates that telecom hardware inventory and engineering planning knowledge continue to provide meaningful predictive capability for site-level power estimation.	MAPE ≈ 4.15% $R^2 \approx 0.9851$
Limitations of cell-wise decomposition-based machine learning The fully data-driven cell-wise ML framework produced the weakest performance among all evaluated approaches. The primary causes of this reduced performance included lack of true cell-level power measurements, synthetic decomposition assumptions and cumulative aggregation error propagation.	MAPE ≈ 11.88% $R^2 \approx 0.9034$
Site-wise ML framework Direct site-level learning significantly improves prediction performance	MAPE ≈ 3.80% $R^2 \approx 0.9888$
Hybrid machine learning framework Combining traffic-aware features, hardware-aware features and infrastructure-related information produced the strongest overall predictive capability	MAPE ≈ 3.74% $R^2 \approx 0.9893$

Table 8-1: Key Findings and Features

Feature importance analysis further confirmed that both traffic demand and installed telecom hardware play major roles in determining site-level power consumption behavior.

Residual analysis also demonstrated that,

- most sites remained within acceptable operational prediction margins
- the final hybrid ML framework produced significantly lower residual variability compared to other methods.

8.3. Research Contributions

This project made several practical and technical contributions to telecommunication infrastructure analytics and power planning.

- The project developed a telecom-specific engineering-rule power prediction framework simulating real-world infrastructure planning practices used within operational telecommunication environments.
- The project implemented and evaluated a fully data-driven cell-wise ML prediction framework capable of performing
  - cell-level prediction, site-level aggregation and decomposition-based power estimation
- The project developed a fully data-driven site-wise ML framework which demonstrated significantly improved prediction accuracy and operational scalability.
- The project proposed a final hybrid machine learning framework combining traffic-aware learning, hardware-aware feature engineering and infrastructure-aware operational analytics. This framework achieved the strongest predictive performance and demonstrated high practical suitability for telecommunication infrastructure planning applications.
- The project established a unified comparative framework for evaluating engineering-rule approaches, cell-wise ML approaches, site-wise ML approaches and hybrid ML methodologies within the same operational telecom dataset environment.

8.4. Final Conclusion

This project demonstrated that data-driven machine learning frameworks can significantly improve telecommunication site power prediction compared to traditional infrastructure planning methodologies.

Although engineering-rule approaches continue to provide strong operational interpretability and reasonable predictive capability, machine learning techniques offer important advantages including:

- adaptive learning
- nonlinear relationship modeling
- scalability
- improved prediction accuracy.

The results clearly demonstrated that site-wise machine learning approaches outperform, cell-wise decomposition-based approaches for telecom and site power prediction.

The final hybrid ML framework achieved the best overall performance by integrating aggregated traffic behavior, hardware inventory features, and infrastructure-aware learning. The achieved performance MAPE ≈ 3.74% and $R^2 \approx 0.9893$ demonstrates excellent operational suitability for supporting,

- rectifier upgrade planning
- battery sizing
- infrastructure capacity assessment
- telecom modernization programs

The project therefore confirms that AI-driven operational analytics can provide substantial value for

- telecommunication infrastructure planning
- power upgrade decision support
- operational risk reduction
- infrastructure investment optimization

Ultimately, the proposed framework demonstrates that data-driven site-level machine learning can become a highly effective decision-support tool for modern telecommunication operators managing increasingly complex multi-technology network environments.

Master of Data Science & Artificial Intelligence

CS5998 | Capstone Project

Appendix A

Reproducibility Checklist

Data-Driven Power System Upgrade Planning for Cellular Network Sites Using Traffic & Hardware Configuration

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1. Appendix A — Reproducibility Checklist

1.1. Reproducibility Checklist

This appendix summarizes the reproducibility aspects of the capstone project titled *"Data-Driven Power System Upgrade Planning for Cellular Network Sites Using Traffic & Hardware Configuration"*. The checklist documents datasets, preprocessing procedures, modeling approaches, evaluation methodology, software environment, and implementation details required to reproduce the reported results. The project structure, methodologies, datasets, and evaluation metrics are described throughout the final report.

Reproducibility Checklist

Category	Item	Status	Description
Project Scope	Problem Definition	YES	Telecommunication site power prediction using traffic and hardware configuration data
	Objectives Clearly Defined	YES	Predict site power demand and compare engineering-rule and ML approaches
	Comparative Methodology	YES	Four different prediction frameworks implemented and evaluated
Datasets	Dataset Sources Documented	YES	LTE traffic, 5G traffic, site power, and site database datasets documented in report
	Dataset Granularity Specified	YES	15-minute operational interval measurements
	Dataset Structure Explained	YES	Site_ID, trigger_ID, datetime, traffic, power, and error fields defined
	Data Integration Process Described	YES	Dataset merging based on Site_ID and trigger_ID
	Missing Value Handling Explained	YES	Missing values replaced using preprocessing techniques
Preprocessing	Data Cleaning Steps	YES	Column standardization, datatype conversion, synchronization
	Feature Engineering Explained	YES	Traffic aggregation, equipment counts, derived telecom features
	Aggregation Logic Documented	YES	Cell-wise and site-wise aggregation procedures described
	Train/Test Methodology	YES	Time-based split methodology used
Modeling	Engineering Rule Framework	YES	Telecom engineering-rule framework implemented
	Cell-Wise ML Framework	YES	Implemented using traffic decomposition and aggregation
	Site-Wise ML Framework	YES	Implemented using aggregated traffic features
	Hybrid ML Framework	YES	Combined traffic and hardware-aware ML framework
	ML Algorithm Selection Justified	YES	Random Forest regression selected and justified

Reproducibility Checklist (Continue)

Category	Item	Status	Description
Implementation	Source Code Provided	YES	Jupyter notebooks submitted
	Notebook-Based Workflow	YES	End-to-end implementation available in notebooks
	Output Files Generated	YES	Prediction outputs and summary outputs exported as Excel files
	Graphical Evaluation Included	YES	Scatter plots, residual plots, histograms, feature importance graphs
Evaluation	MAE Reported	YES	Included in comparative evaluation
	RMSE Reported	YES	Included in comparative evaluation
	MAPE Reported	YES	Included in comparative evaluation
	R^2 Reported	YES	Included in comparative evaluation
	Residual Error Analysis	YES	Site-wise residual and absolute error analysis performed
	Comparative Analysis	YES	All four methods evaluated using same dataset environment
Output	Prediction Excel Outputs	YES	Output prediction datasets generated
	Summary Datasets	YES	Site-wise summary datasets generated
	Site-Wise Error Analysis	YES	Implemented
	Feature Importance Analysis	YES	Implemented for ML frameworks
Software Environment	Programming Language	YES	Python
	Development Environment	YES	Jupyter Notebook
	Main Libraries Documented	YES	Pandas, NumPy, Scikit-learn, Matplotlib
Operational Reproducibility	Workflow Description	YES	End-to-end workflow documented in methodology chapter
	Engineering Assumptions Explained	YES	Telecom hardware assumptions documented
	ML Evaluation Strategy Explained	YES	Regression metrics and residual analysis documented
Limitations	Dataset Constraints Discussed	YES	Limited operational duration and available measurements
	Cell-Level Power Measurement Limitation	YES	Explicitly discussed in report
	Generalization Constraints	YES	Discussed under project limitations
Deliverables	Final Report	YES	Submitted
	Source Code	YES	Submitted as notebooks
	Reproducibility Checklist	YES	Included as Appendix A

1.2. Submitted Source Code Files

The following implementation notebooks were submitted as part of the project:

#	File Name	Purpose
1	site_power_ML_cell_to_site.ipynb Github Link	Machine learning framework developed to estimate cell-level power consumption using traffic information before aggregating predictions to site level. This approach enabled evaluation of decomposition-based telecom power prediction. ***Cell wise power prediction and aggregate to predict the final power of a site***
2	site_power_ML_site_to_site.ipynb Github Link	Machine learning framework developed using aggregated site-level traffic directly as model input. This approach significantly improved prediction accuracy by reducing aggregation-related error propagation. ***Aggregate the Traffic of all cells of a site and predict a power based on Traffic***
3	2G_3G_Eng_Rule.ipynb Github Link	Predict power contribution from 2G and 3G based on industry known power parameters and integration with the site inventory.
4	LTE_Eng_Rule.ipynb Github Link	Prediction of power based on LTE traffic, based on industry known parameters.
5	NR_Eng_Rule.ipynb Github Link	Prediction of power based on 5G NR traffic, based on industry known parameters.
6	Site_Power_Prection_Eng_Rule.ipynb Github Link	Prediction of Total power of a site based on 2G, 3G, 4G and 5G NR power (3 + 4 + 5)
7	site_power_ML.ipynb Github Link	Machine learning framework combining traffic information and hardware configuration features developed using Random Forest regression. This model achieved the best overall performance among all evaluated methods.

1.3. Output Files

#	Output File	Framework
1	Fully_Data_Driven_Cell_Wise_Predictions.xlsx	site_power_ML_cell_to_site.ipynb
2	Fully_Data_Driven_Site_Wise_Predictions.xlsx	site_power_ML_site_to_site.ipynb
3	Fully_Data_Driven_Cell_Wise_Predictions_Summary.xlsx	site_power_ML_cell_to_site.ipynb
4	Fully_Data_Driven_Site_Wise_Predictions_Summary.xlsx	site_power_ML_site_to_site.ipynb
5	site_power_2g3g.xlsx	Site_Power_Prection_Eng_Rule.ipynb
6	sec_power_lte.xlsx	Site_Power_Prection_Eng_Rule.ipynb
7	site_power_lte.xlsx	Site_Power_Prection_Eng_Rule.ipynb
8	sec_power_5g.xlsx	Site_Power_Prection_Eng_Rule.ipynb
9	site_power_5g.xlsx	Site_Power_Prection_Eng_Rule.ipynb
10	Total_Power_Prediction_Eng_Rule.xlsx	Site_Power_Prection_Eng_Rule.ipynb
11	Total_Power_Prediction_Eng_Rule_Summary.xlsx	Site_Power_Prection_Eng_Rule.ipynb
12	Final_Site_Power_Predictions.xlsx	site_power_ML.ipynb
13	Final_Site_Power_Predictions_Summary.xlsx	site_power_ML.ipynb

1.4. Software Environment

Component	Technology
Programming language	Python
Development platform	Jupiter notebook, Google Colab
Source code repository	GitHub used for dataset hosting and notebook management
Data processing	Pandas, NumPy
Machine learning	Scikit-learn
Visualization	Matplotlib
ML algorithm	Random forest regression
Cloud Execution Environment	Google Colab is used for notebook execution and experimentation
Dataset management tool	Microsoft excel used for telecom operational datasets
Architecture Design draw Tool	Draw.io used for system architecture and workflow diagrams
Report preparation tool	Microsoft Word

The project implementation and experimentation were primarily conducted using Google Colab notebooks. Source datasets and notebook synchronization were managed through GitHub repositories to improve reproducibility, centralized storage, and remote accessibility of project resources.

Operational datasets were primarily maintained and processed using Microsoft Excel files before preprocessing within Python-based analytical workflows. System architecture diagrams and workflow visualizations were created using draw.io to support methodology explanation and framework visualization within the final report.

1.5. Reproduction Procedure

To reproduce the experimental results:

1. Prepare the operational datasets described in Chapter 4.
2. Execute preprocessing and feature engineering steps.
3. Run the engineering-rule notebooks for technology-specific power estimation.
4. Execute the ML notebooks for:
   - Cell-wise ML prediction
   - Site-wise ML prediction
   - Hybrid ML prediction
5. Generate prediction outputs and summary datasets.
6. Execute evaluation sections to reproduce:
   - MAE
   - RMSE
   - MAPE
   - R^2
   - Scatter plots
   - Residual analysis
   - Feature importance analysis
7. Compare results using the comparative evaluation framework described in Chapter 6.

1.6. Notes on Reproducibility

Certain operational datasets used in this project originate from internal telecommunication systems of Cable & Wireless Seychelles (C&WS) and may contain commercially sensitive information. Therefore, raw datasets are not publicly distributed. However, the preprocessing logic, modeling workflow, feature engineering methodology, and evaluation framework are fully documented within the report and accompanying notebooks, enabling methodological reproducibility.